

S-JEPA for gait

Telling normal, MS, and Parkinson's apart from a walking video

A hands-on tutorial series that learns motion without labels, then classifies three health conditions and compares against a classical baseline.

The question

Given a short video of someone walking, can we tell whether their gait is:

- **normal**,
- affected by **multiple sclerosis (MS)**, or
- affected by **Parkinson's disease (PD)**?

We try two approaches and compare them fairly:

1. a **self-supervised** model, S-JEPA, that learns motion features with no labels, then
2. a **classical Random Forest** on hand-made gait features (the exp5 recipe).

The data, counted honestly

Condition	Clips	Independent sources
normal	19	16
ms	11	11
pd	17	8
total	47 usable	~35

Some clips are pieces of one longer video. That matters: two clips of one walk are not independent, so we must split by **source**, never by clip.

The pipeline

Both approaches share one pose front-end, then split into two branches, then meet again for a fair comparison on the same test videos.

Step 1: video to skeleton

MediaPipe BlazePose gives **33 landmarks** per frame. Each video becomes an array of shape `(T, 33, 3)`: frames, joints, and `[x, y, visibility]`.

We normalize each frame by centering on the pelvis and scaling by the torso length, so the model sees the **shape of the motion**, not where the person stood or how close the camera was.

No bounding boxes and no annotation files are needed, unlike the GAVD pipeline. We just run pose on the whole frame.

Step 2: tokens

Group $l = 4$ adjacent frames of one joint into a token. With a 32-frame window and 33 joints that is $8 \times 33 = 264$ tokens. Each token also gets a spatial and a temporal position embedding.

How S-JEPA learns

Hide some joints. Predict their **features** (not their coordinates) from the rest. The answers come from a slow **teacher** encoder, an exponential moving average of the student. The teacher is what stops the model from collapsing to a constant.

The masking choice

The paper hides the joints that move the **most**. For clinical gait that is backwards, because the sign is often **less** motion: short steps, stiff knees, reduced arm swing.

So we hide a **fixed set of twelve neurologically relevant joints**: both shoulders and both complete legs. No other joints. Ever.

The masked joints

Idx	Joint	Idx	Joint
11	LEFT_SHOULDER	12	RIGHT_SHOULDER
23	LEFT_HIP	24	RIGHT_HIP
25	LEFT_KNEE	26	RIGHT_KNEE
27	LEFT_ANKLE	28	RIGHT_ANKLE
29	LEFT_HEEL	30	RIGHT_HEEL
31	LEFT_FOOT_INDEX	32	RIGHT_FOOT_INDEX

Twelve joints, chosen once, used every step. This replaces motion-aware masking entirely.

Keeping the features from collapsing

The paper uses three tools: a slow EMA teacher, centering, and sharpening. Remove the teacher and the model collapses to near chance.

We add a fourth, **VICReg**, clearly labeled as an extension. It has three terms:

- **variance**: keep each feature dimension spread out,
- **invariance**: two views of one window should match,
- **covariance**: decorrelate the dimensions.

We use it to pull the three condition clusters apart.

VICReg, visually

Progressive training

Learn ordinary walking first, then broaden the model's world with MS and PD, then add VICReg to separate the classes.

Fair evaluation: split by source

All clips of one source stay together. We report **grouped k-fold** with mean and standard deviation, because a single split of a few dozen videos is too noisy to trust.

Results (grouped k-fold, laptop profile)

Metric	Random Forest	S-JEPA probe
accuracy	~0.66 +/- 0.09	~0.57 +/- 0.10
macro F1	~0.67 +/- 0.10	~0.57 +/- 0.11

On a few dozen fully labeled videos the classical baseline is strong, exactly as expected. Numbers move with the profile and the random seed.

What this does and does not show

Fair. Both models saw identical, leakage-safe folds and identical metrics.

Honest. With ~35 independent sources, a few videos moving between folds swings the score several points. This is a methodology demonstration, not a clinical result.

Where S-JEPA earns its keep. Its features come from unlabeled motion, so its edge shows up when labels are scarce or when it is pretrained on far more walking than we have here. The label-efficiency sweep hints at this.

Try it yourself

- Local: `cd experiments/multiple-sclerosis && uv sync && uv run jupyter lab`
- Colab: click the badge at the top of any notebook
- Switch model size with `SJEPA_PROFILE=laptop` or `gpu` in `.env`
- Fast check: `SJEPA_SMOKE=1 uv run python sjepa/tests/test_smoke.py`

Seven notebooks take you from raw video to a fair, reproducible comparison.

Thank you.